GHOST INFRASTRUCTURE

How 19th-Century Coal Geography Still Shapes Who Gets a "15-Minute Life" Today — Now Tested Across Two Cities and Three Thresholds

Executive Summary · v1.1.0 · DOI: 10.5281/zenodo.21761320 · Sakshi D. Maske

The Question

Bochum's modern shape was never designed for people — it was built around coal mines and steel works, with railways, roads, and worker-housing colonies (Zechensiedlungen) laid out to serve 19th-century industry. Coal mining ended in Bochum in 1974. More than half a century later, does that historical industrial geography still leave a measurable imprint on which neighborhoods get a genuine "15-minute city" — walkable access to essential services — and which don't? And does any such effect hold up beyond one city and one arbitrary walking-time cutoff?

The Method

13 historical coal mines and 4 worker-housing colonies were digitized for Bochum from Mindat.org and German heritage archives. A true network-based 15-minute accessibility model was built across Bochum's complete 69,393-node pedestrian street network (OSMnx) — not a simplified straight-line radius. Historical-site proximity was statistically tested against present-day accessibility via a Welch's t-test, then verified against its most obvious confound (city-center proximity) using correlation and logistic regression, and independently corroborated with a Local Moran's I spatial-clustering analysis.

The Finding

A reversed, independently-verified effect. The hypothesis was that historical industrial sites would predict present-day neglect. The evidence showed the opposite: low-accessibility zones are, on average, further from historical industrial sites than high-accessibility zones — a highly significant, medium-to-large effect that holds independently of city-center proximity.

Metric	Value
Low-accessibility nodes — avg. distance to site	1,984 m (n = 9,858)
High-accessibility nodes — avg. distance to site	1,450 m (n = 59,535)
Welch's t-test	t = 42.887, $p < 0.00001$ (Cohen's d = 0.589)
Confound check — distance-to-center correlation	r = 0.063 (genuinely independent)
Logistic regression (controlling for center)	coefficient = -0.0005, $p < 0.001$
Local Moran's I corroboration	97.1% of low-access nodes in significant cold-spot clusters

> Interpretation: 19th-century industrial cores were built dense, by necessity, around the mines and colonies where workers actually lived — that density appears to persist as present-day street connectivity and service coverage. A "path dependency of centrality," not the originally hypothesized "path dependency of neglect."

Two Robustness Extensions NEW

1. Is 15 minutes a special threshold? The full pipeline was re-run at a stricter 10-minute (750m) and a more permissive 20-minute (1,500m) cutoff. The reversed effect holds — and actually strengthens — at every threshold tested (Cohen's d = 0.413 at 10-min, 0.589 at 15-min, 0.661 at 20-min).

2. Does it hold in a second city? The identical methodology was independently replicated in Essen (4 mines + 4 worker colonies, 72,027-node network). The raw reversed effect and the spatial-clustering result both replicate (t = 24.731, $p < 0.00001$, Cohen's d = 0.338; 95.5% of low-access nodes in cold-spot clusters). The confound-independence result does *not* replicate: in Essen, historical-site distance correlates moderately with city-center distance (r = 0.405 vs. Bochum's r = 0.063), so the "independent of city center" claim specifically should be read as Bochum-specific, not yet general across the Ruhr Valley. This mixed result is reported in full — including the part that didn't replicate — rather than only the part that confirms the original city's finding.

Validation & Robustness Checklist

- ✓ Full network-based accessibility model (69,393 nodes in Bochum, 72,027 in Essen — not a straight-line radius)
- ✓ Most obvious confound explicitly tested (city-center proximity, r = 0.063 in Bochum)
- ✓ Logistic regression confirms independence in Bochum ($p < 0.001$, controlling for center distance)
- ✓ Independent corroboration via Local Moran's I spatial clustering (KNN k=8, 99 permutations) — in both cities
- ✓ Effect size reported, not just significance (Cohen's d = 0.589 Bochum, 0.338 Essen)
- ✓ Effect holds — and strengthens — across 10-, 15-, and 20-minute walking thresholds
- ✓ Independently replicated in a second city (Essen), with the part that didn't fully replicate reported honestly
- ✓ Reversed-hypothesis result reported honestly throughout, not reframed to fit the original prediction

Honest Limitation

This finding is now verified within Bochum *and* independently replicated in a second Ruhr Valley city, Essen — but the specific claim that the effect is statistically independent of city-center proximity did not replicate in Essen, so that narrower claim should be treated as Bochum-specific rather than a general Ruhr Valley finding pending a larger Essen dataset. The relationship throughout remains correlational, not causal: the confound check rules out city-center proximity specifically, but cannot rule out every possible alternative explanation. Historical mine and colony locations were digitized from secondary sources (Mindat.org, KuLaDig, Wikipedia, regional heritage archives) rather than primary survey records, and Essen's 8-site dataset is a smaller fraction of that city's full historical mining register than Bochum's 17-site dataset is of Bochum's.

Real-World Relevance

Urban planners often assume former industrial zones are the neighborhoods most likely to be left behind — under-served and poorly connected. This project tested that assumption directly, on real network data across two cities and three accessibility thresholds, and found the opposite in the raw effect and spatial clustering every time. Getting this backwards has real planning consequences: hundreds of European cities share Bochum's and Essen's 19th-century industrial-core urban form, and directing accessibility investment toward the wrong neighborhoods based on an intuitive but incorrect assumption is a real, avoidable cost.

GitHub: github.com/sakshimaske303-commits/GHOST_INFRASTRUCTURE | Live Dashboard: ghostinfrastructure-areytvp4x8ofu6l5tosj2z.streamlit.app | Zenodo DOI: 10.5281/zenodo.21761320
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